Class 5: Data Viz with ggplot

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Background

There are many *graphics systems* in R for making plots and figures. These include **"base R graphics"** like the `plot()` and add-on packages like **ggplot2**.

Let's compare how we make a simple graph with these two systems:

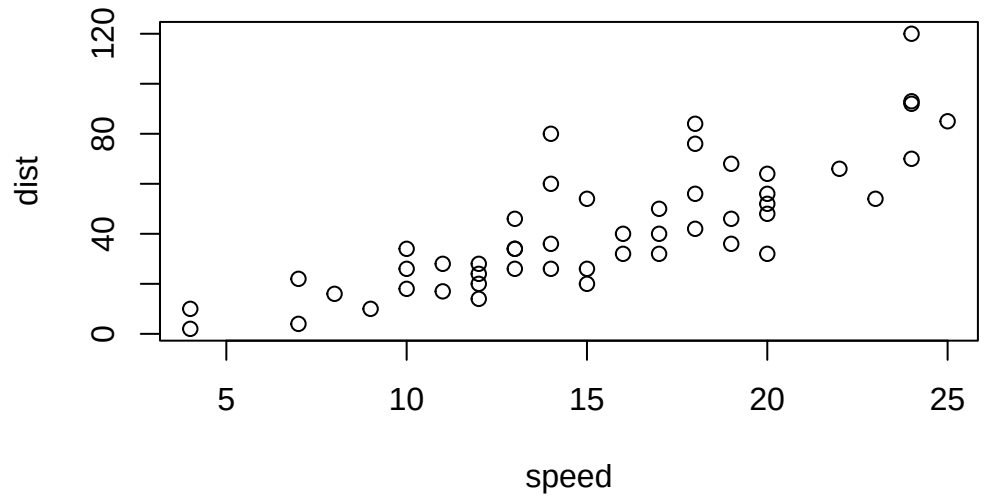
We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

A base R plot:

```
plot(cars)
```



Now for **ggplot2**. before I can use ggplot2 I need to install it on my computer. To do this we can use the function`install.packages("ggplot2")`

N.B. We never run an `install.packages()` in our quarto doc (we run it in our R console) as it would re-install the package every time I render my document.

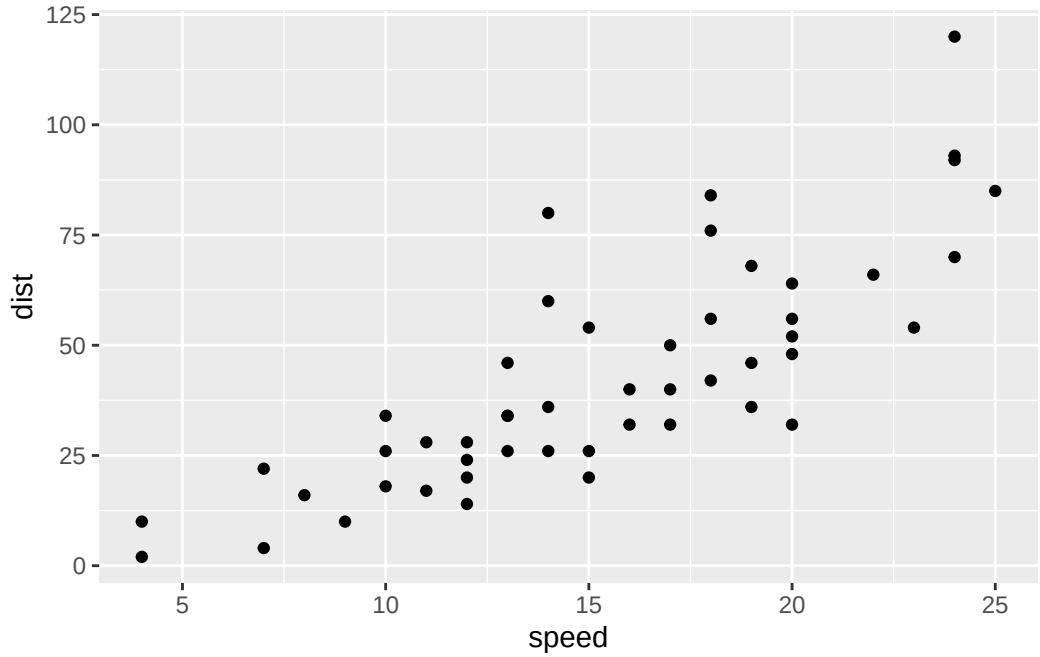
Once installed we still need to load up the packages into our R session. We do this with a `library()` call

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 inputs:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot),
- the **geometry** layer (how you want the plot drawn, points, lines, etc.)

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```

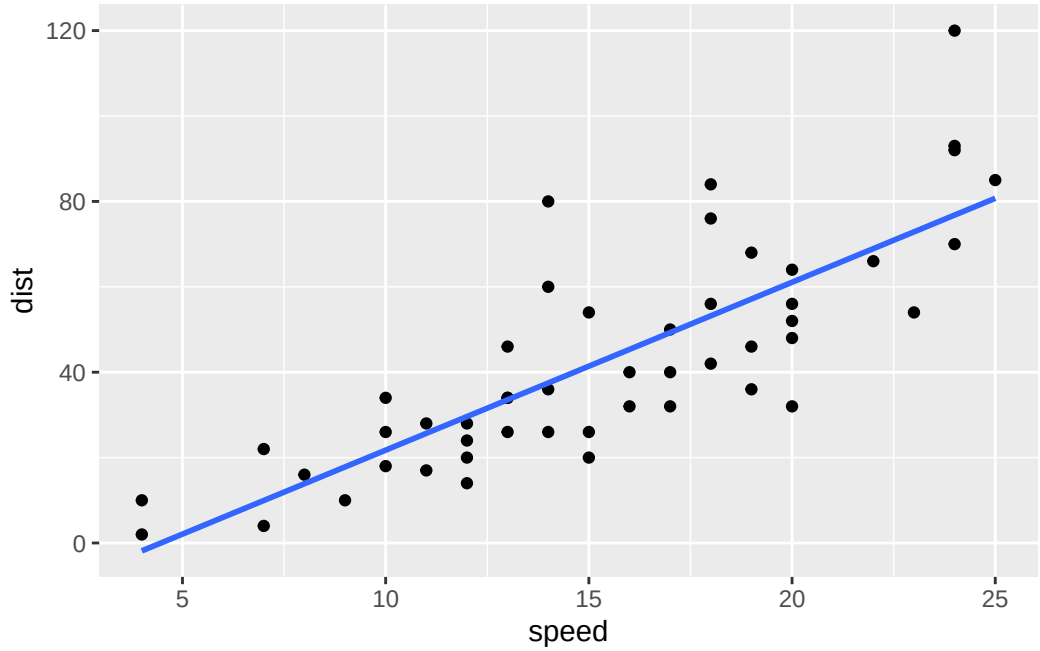


Let's add some features (more layers)

Let's add a trend line that shows the relationship between `speed` and `distance`:

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

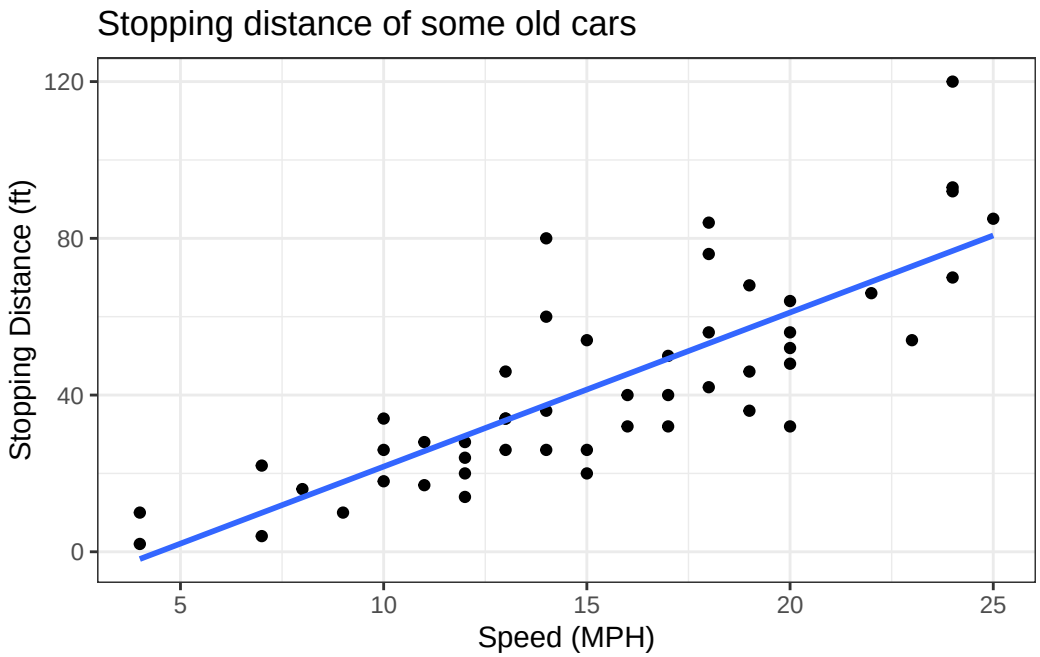
```
`geom_smooth()` using formula = 'y ~ x'
```



Let's add some more details

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(x="Speed (MPH)", y="Stopping Distance (ft)", title="Stopping distance of some old cars
```

```
`geom_smooth()` using formula = 'y ~ x'
```



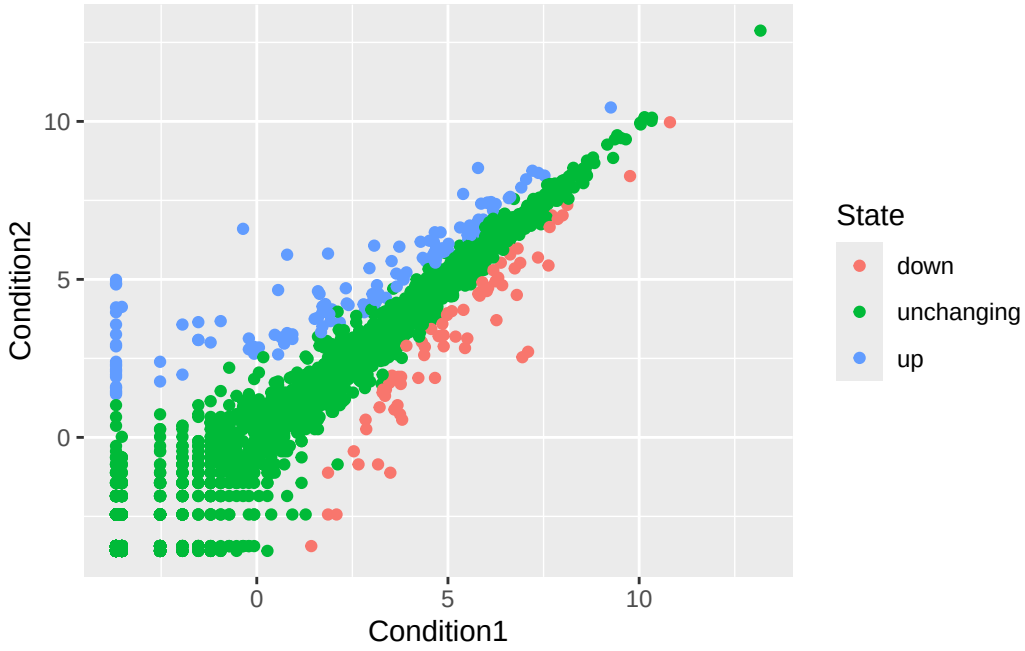
Gene Expression Plot

First we will read some expression data for the effects of a GLP-1 drug

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
```




```
nrow(genes)
```

```
[1] 5196
```

```
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
n.total <- nrow(genes)
n.up <- sum(genes$State=="up")

round(n.up/n.total*100, 2)
```

```
[1] 2.44
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual(values=c("pink","lightblue","purple")) +
  theme_bw() +
  labs(x="No drug", y="Drug", title="GLP-1 inhibitor effects on gene expression")
```



Going further

Here we read the famous *gapminder*

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different "country" values are in this dataset>

```
length(table(gapminder$country))
```

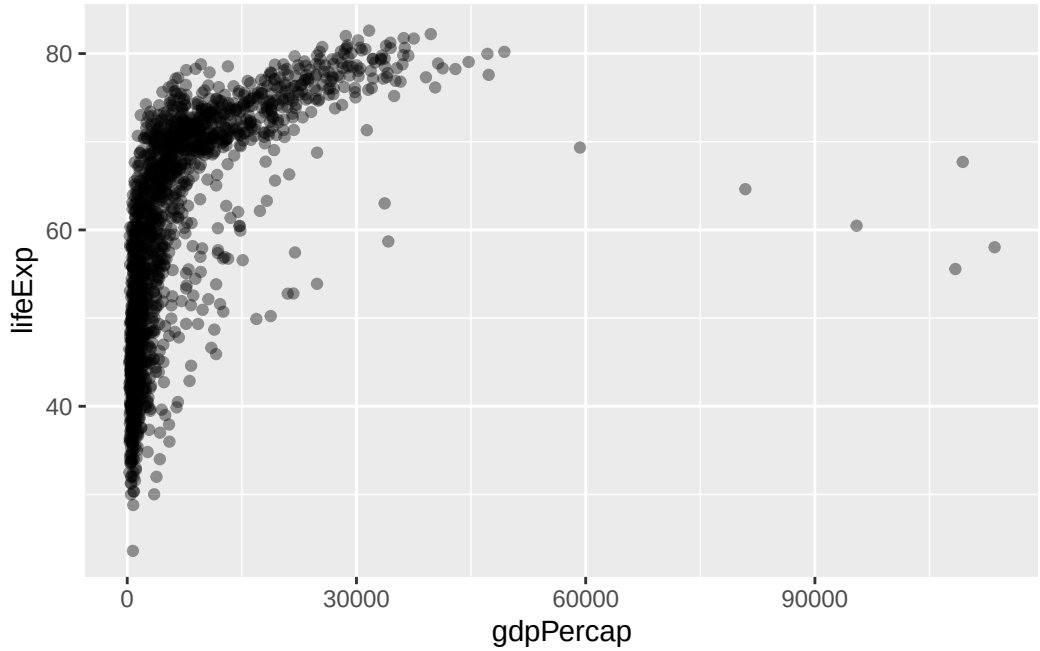
```
[1] 142
```

```
length(unique(gapminder$country))
```

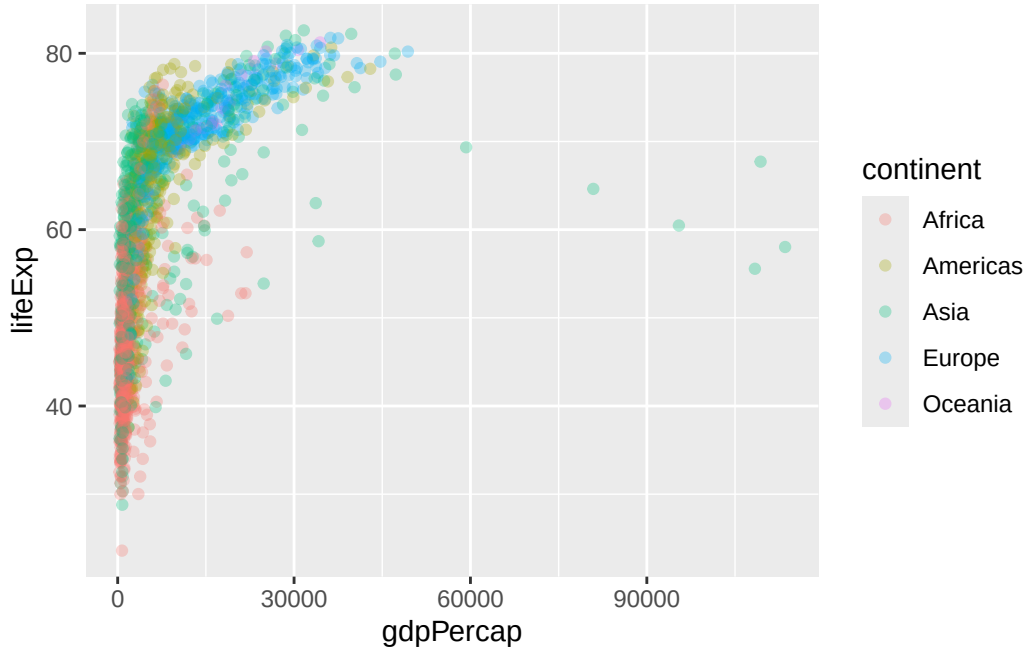
```
[1] 142
```

let's make our first plot of `gdpPercap` vs `lifeExp`:

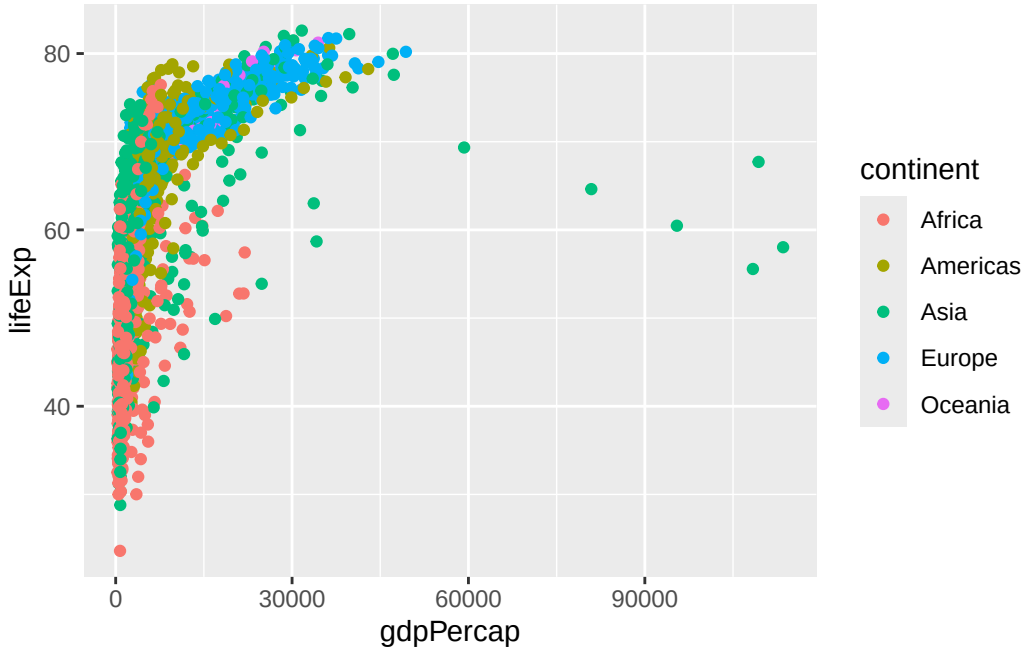
```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.4)
```




```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point(alpha=0.3)
```

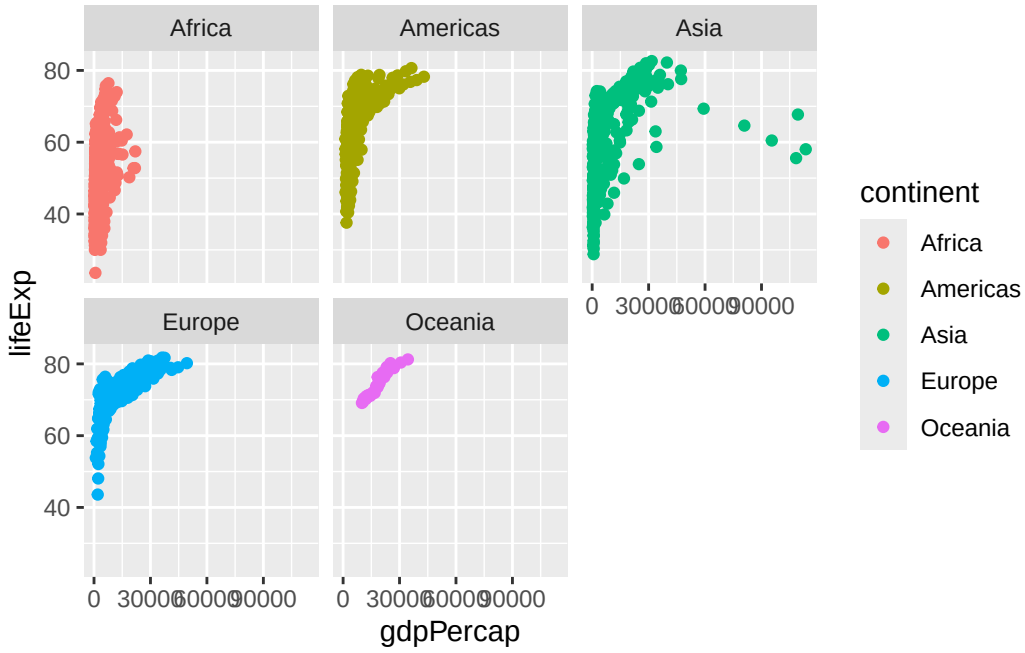


```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point()
```



One useful layer we can add to explore big data sets like this is called "facetting".

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point() +
  facet_wrap(~continent)
```



Limit to 1977 and 2007

We can install and use the **dplyr** package to help is "filter" the full dataset down to a smaller subset that only includes the years we want.

First we need to run `install.packages("dplyr")` and then load it with `library(dplyr)`.

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year==2007, country=="United States")
```

```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 2007  78.242 301139947  42951.65
```

```
filter(gapminder, year==1977, country=="United States")
```

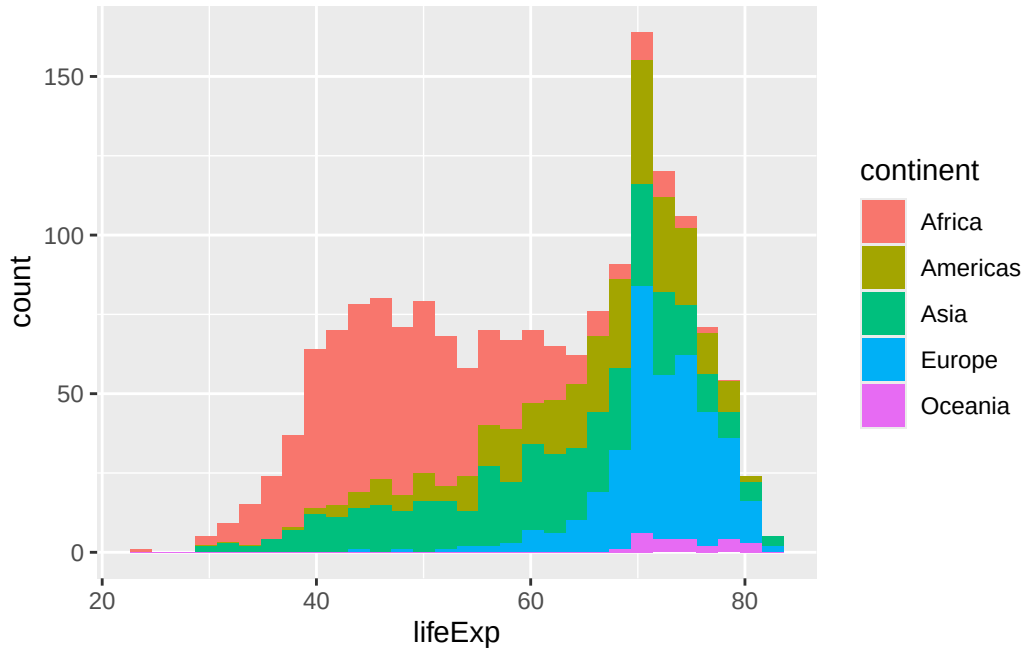
```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 1977   73.38 220239000  24072.63
```

```
#gapminder$year
```

Q. Make a Histogram of lifeExp fill color by continent.

```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_boxplot()
```

